

Badr Albdulmajeed

+971502859094 | bbhg.work@gmail.com | UAE (Valid Golden Visa)

LinkedIn: <https://www.linkedin.com/in/badr-albdulmajeed-b45437155/>

ABOUT ME

I am a fresh Electrical and Electronic Engineering graduate with Honours from Asia Pacific University of Technology & Innovation (APU). During university and my R&D role at Control Easy, I worked on power systems, protection, PLCs, sensors, embedded hardware, industrial automation, and computer vision. My final year project focused on PCB defect detection using a self-supervised Vision Transformer. I am currently in the UAE with a valid Golden Visa and am available for a full-time engineering role.

SKILLS AND STRENGTHS

- **Power Systems and Protection:** Load flow analysis, overcurrent relay coordination, IDMT relays, AC microgrid protection, MATLAB/Simulink, and PowerWorld.
- **PLC and Industrial Automation:** Siemens S7-1200, S7-300, and S7-200 PLCs, TIA Portal, SCL, 24 V sensors, relay logic, pneumatic circuits, and Industry 4.0 equipment.
- **Embedded Systems:** Raspberry Pi 5, NVIDIA Jetson Orin Nano, Arduino, ESP32, GPIO, sensor interfacing, camera integration, and optical and relay isolation.
- **Computer Vision and AI:** Python, OpenCV, PyTorch, Vision Transformers, YOLOv8, DINOv2, RF-DETR, ONNX Runtime GPU, Roboflow, classification, detection, and segmentation.
- **Engineering Software:** SolidWorks, TinkerCAD, LTspice, DIALux, LabVIEW, MATLAB App Designer, and ROS2.
- **Programming:** Python, C, C++, C#, MATLAB, and LabVIEW.
- **Working style:** I take care with details, communicate clearly, and learn new tools by testing them and documenting what I find.

EXPERIENCE

R&D Project Engineer | Control Easy Seri Kembangan, Malaysia

March 2025 - July 2025

- Developed and tested machine-vision applications for industrial quality control using classification, object detection, and segmentation.
- **Automated inspection:** Built a golf-ball inspection station using a Raspberry Pi 5, proximity sensor, camera, and Vision Transformer model for real-time classification and sorting.
- **Hardware interfacing:** Connected a 24 V industrial proximity sensor to 3.3 V Raspberry Pi GPIO using optical and relay isolation, then tested the complete hardware and software setup.
- **PLC integration:** Integrated vision systems with PLC-controlled Industry 4.0 equipment and received hands-on training with Siemens S7-1200, S7-300, and S7-200 PLCs using SCL in TIA Portal.

- **Technical documentation:** Prepared manuals covering dataset creation, model training, GUI deployment, hardware setup, and electrical wiring.

SELECTED PROJECTS

1. **Final Year Project - Self-Supervised Vision Transformer for PCB Defect Detection:** Built a PCB inspection system with PyTorch, SimCLR, MaxViT-Small, RF-DETR, DINOv2, ONNX Runtime CUDA, Streamlit, OpenCV, and Roboflow. Achieved 77.50% test accuracy, 0.8281 weighted F1 across 22 classes, 68.83% mAP@50, and 37.25 FPS after GPU optimisation.
2. **Hospital Logistics AGV Vision System:** Developed the vision subsystem for an autonomous hospital logistics robot using NVIDIA Jetson Orin Nano, ZED 2i, YOLOv8, PyTorch, ONNX Runtime GPU, ROS2, PyQt5, and Roboflow. Achieved 98.99% mAP@50, 98.66% precision, and 97.85% recall.
3. **Overcurrent Relay Coordination Tool:** Created a MATLAB App Designer tool for IDMT relay coordination, operating-time calculations, Time-Current Characteristic curves, system analysis, and validation reporting for ten overcurrent relays. Maximum calculation error: 1.00%.
4. **10-Bus Power Flow Analysis:** Built and analysed a 10-bus system using Newton-Raphson, Gauss-Seidel, and Fast Decoupled load-flow methods. Evaluated bus voltage, line flow, generation, and losses under additional loading.
5. **AC Microgrid Overcurrent Protection:** Modelled a 3-bus AC microgrid in MATLAB/Simulink with transformers, distribution lines, CTs, circuit breakers, and overcurrent relays. Tested IEC and IEEE inverse relay characteristics under different fault types.

OTHER PROJECTS

- **Medication Scanner and Obstacle Detection GUI (Python, Jetson):** Developed a Python GUI on Jetson for medication image scanning and obstacle detection.
- **Modular Smart Power Hub - The Honeycomb System (SolidWorks):** Designed a modular power distribution hub in SolidWorks with a honeycomb-inspired structure.
- **Automated Golf Ball Quality Control System (Raspberry Pi, Vision Transformer):** Built a Raspberry Pi inspection setup using camera input and a Vision Transformer to classify golf balls.
- **PCB Component Inspection System (OpenCV, AI):** Used OpenCV and AI-based image processing to inspect PCB components and identify visual defects.
- **Wood Defect Segmentation System (Computer Vision):** Applied computer-vision segmentation to locate and classify visible defects in wood samples.
- **Fabric Quality Control System (Computer Vision):** Created a computer-vision workflow to inspect fabric images and identify quality issues.
- **Vaccination Center System (Python):** Developed a Python system to organise vaccination-centre records and basic service flow.

- **Vaccination Inventory System (C++):** Built a C++ inventory system to track vaccine stock, records, and updates.
- **Coffee Making Process (LabVIEW):** Created a LabVIEW process model for monitoring and controlling coffee preparation steps.
- **Automated Bus Door Control (PLC and Pneumatics):** Designed a PLC and pneumatic control sequence for safe automated bus-door operation.
- **Automotive Part Control System (C Programming):** Programmed a C-based control system for handling and checking automotive parts.
- **Library Lighting Design (DIALux):** Planned library lighting levels and fixture layout using DIALux.
- **Automated Parking System (Arduino):** Built an Arduino-based parking system for vehicle detection and space control.
- **Sound Purifier (MATLAB):** Developed a MATLAB signal-processing model to reduce unwanted sound.
- **BJT Audio Amplifier (MATLAB):** Simulated and analysed a BJT amplifier for audio signal amplification in MATLAB.
- **Discrete-Time System using Z-Transform:** Analysed a discrete-time system using Z-transform methods and MATLAB calculations.
- **Temperature Control System (Arduino and TinkerCAD):** Built an Arduino and TinkerCAD model for temperature sensing and automatic control.
- **AI Hand Gesture Recognition (Python and OpenCV):** Developed a Python and OpenCV system to recognise hand gestures from camera input.
- **Vision-Based Fruit Inspection System (Python, OpenCV, Roboflow):** Used Python, OpenCV, and Roboflow to inspect fruit images for visual quality classification.
- **Single-Phase Transformer and Three-Phase Alternator Simulation:** Simulated transformer and alternator behaviour and reviewed voltage and current characteristics.
- **FM Modulation and Demodulation (LabVIEW):** Built a LabVIEW model for FM signal generation, modulation, and demodulation.
- **Microgrid Technology Simulation (MATLAB and LabVIEW):** Simulated microgrid operating concepts in MATLAB and LabVIEW.
- **Off-Grid Wind Energy System (Simulink):** Modelled an off-grid wind energy system and reviewed its output in Simulink.
- **Go-Kart Design (SolidWorks):** Designed a go-kart assembly and mechanical components in SolidWorks.

EDUCATION

- **Asia Pacific University of Technology & Innovation (APU):** Bachelor of Electrical and Electronic Engineering with Honours | 2020-2022; withdrew; rejoined 2024-2026 | Completed 2026 | Fresh Graduate
- **Sama American School:** American High School Diploma | Cairo, Egypt | 2017-2019

Courses & Certificates

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| • MATLAB Onramp | MathWorks (Sep 2026) |
| • Simulink Onramp | MathWorks (Sep 2026) |
| • Core MATLAB Skills Learning Path (4 courses: MATLAB Desktop Tools & Troubleshooting Scripts, Explore Data with MATLAB Plots, Make and Manipulate Matrices, Calculations with Vectors and Matrices) | MathWorks (Sep 2026) |